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Thermochemical methanolysis-based recyclability of polyethylene terephthalate (PET) via pseudo-catalytic depolymerisation

Dohee Kwon [a,1], Doyeon Lee [b,1], Jechan Lee [c], Jee Young Kim [a,d], Eilhann E. Kwon [a,*]

[a] Department of Earth Resources and Environmental Engineering, Hanyang University, Seoul, 04763, Republic of Korea
[b] Department of Civil and Environmental Engineering, Hanbat National University, Daejeon, 34158, Republic of Korea
[c] Department of Global Smart City & School of Civil, Architectural Engineering, and Landscape Architecture, Sungkyunkwan University, Suwon, 16419, Republic of Korea
[d] Department of Environmental and Safety Engineering, Ajou University, Suwon, 16499, Republic of Korea

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ABSTRACT

The global production and consumption of polyethylene terephthalate (PET) have continued to rise, accounting for approximately one-quarter of total plastic output. Currently, PET is derived from fossil-based resources, and the majority of used PET follows a linear, single-use trajectory, ultimately contributing to environmental burdens. While PET has contributed to modern convenience, the alarming statistic that nearly one million PET bottles are discarded every minute emphasises the urgent need to reassess the sustainability of its current production and disposal practices. In this study, a catalyst-free depolymerisation technology, termed pseudo-catalytic transesterification, was developed as a sustainable valorisation approach for the recovery of resources from PET. This approach enabled rapid depolymerisation, achieving high yields of dimethyl terephthalate (DMT, 75.0 %) and ethylene glycol (EG, 17.0 %) within 1 min at 400 °C. Dimethyl carbonate (DMC) was introduced as a (co) solvent to enhance monomer recovery. A 1:1 M ratio of methanol:DMC yielded the highest DMT and EG recoveries outperforming methanol alone. However, excessive DMC promoted ethylene carbonate (EC) formation, thereby reducing the EG yield. Under optimised conditions, the process achieved a recycled PET (rPET) conversion of 68 % with an estimated economic value of $4392 per tonne of rPET. Although the production cost remains higher than that of fossil-based PET, this approach offers substantial environmental benefits by reducing 2 tonnes of CO_2 emissions per tonne of recycled PET waste.

1. Introduction

Polyethylene terephthalate (PET), a thermoplastic polyester, is extensively used in packaging, automotive, and textile industries owing to its durability, transparency, and water resistance [1,2]. In 2021, global PET production reached 61 million tonnes, representing 25 wt % of total plastic output [3]. Its widespread use has led to substantial environmental challenges, with over one million PET bottles sold per minute and more than 90 % discarded in landfills or oceans [4,5]. Owing to its chemical stability, PET persists for 450 years before degrading naturally [6]. Under environmental conditions, PET fragments into macro- and micro-plastics posing risks to marine ecosystems [7]. Moreover, PET production is also resource-intensive, requiring 70–80 MJ kg^{-1} and emitting 2.44 kg CO_2 kg^{-1} [8]. Thus, recycling PET waste is critical for mitigating the environmental impacts and fostering a circular economy [9].

Currently, mechanical recycling dominates PET waste management [10]. This process involves the removal of contaminants and PET reprocessing via melt extrusion [11]. However, mechanical recycling degrades polymer quality and performance over successive cycles, leading to downcycling [12,13]. Thus, chemical recycling has gained attention because of its potential to enhance plastic circularity by depolymerising PET into monomers and value-added chemicals [14,15]. PET is amenable to chemical recycling because of the reversible nature of the esterification and transesterification reactions [16]. This reversibility enables solvent-mediated cleavage of ester linkages to yield monomers [17]. Various chemical recycling pathways have been developed including glycolysis in ethylene glycol [4], methanolysis in

* Corresponding author.
E-mail address: ek2148@hanyang.ac.kr (E.E. Kwon).
[1] These authors contributed equally to this work.

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methanol [18], and alkaline hydrolysis [19].

Among the chemical recycling methods, methanolysis is the most widely used, and is a transesterification reaction that involves the depolymerisation of PET waste using methanol at elevated temperatures (180–280 °C) and pressures (20–40 atm), yielding dimethyl terephthalate (DMT) and ethylene glycol (EG) [20,21]. The produced DMT is chemically equivalent to its virgin counterpart, allowing seamless integration into existing polymer production lines [17,22]. Methanolysis enables efficient purification and is effective for processing low-quality feedstocks, making it a versatile recycling strategy [23]. Catalytic methanolysis has been extensively studied using homogeneous (e.g. metal acetates and ionic liquids) [9,24] and heterogeneous (e.g. metal oxides and metal–organic frameworks) catalysts [25–27]. These catalysts interact with the oxygen atoms in the carboxylic groups of PET, enhancing methanol penetration and accelerating depolymerisation [28]. These catalysts often leave residual contaminants, thereby compromising product purity [9,29], while heterogeneous catalysts suffer from limited dispersibility and large particle sizes, which restrict mass transfer efficiency [30]. Supercritical methanolysis has been explored as a catalyst-free alternative [31,32]. This approach achieves high DMT yields (~98 %) but requires prolonged reaction times (30–40 min) under high pressure conditions (8.26–14.7 MPa) at 300–310 °C, resulting in high capital costs that hinder commercial viability.

This study aimed to develop an economically viable catalyst-free depolymerisation process. A novel transesterification approach, termed pseudo-catalytic transesterification, was used, in which thermal activation is facilitated by a catalytically inert porous material such as silica (Fig. 1) [33,34]. During the reaction, heating PET and methanol generates two distinct phases (molten PET and gaseous methanol) owing to the low boiling point of methanol (65 °C). These reactants are expected to collide within the pores of the medium, inducing a catalytic-like effect that reduces the transesterification activation energy. To further enhance monomer production, we investigated leveraging the high electrophilicity of dimethyl carbonate (DMC) and replaced methanol with DMC [35]. DMC has demonstrated superior reactivity in transesterification, particularly in biodiesel production [33,36], and has been used as a trapping agent for EG in the catalytic methanolysis of PET, shifting the equilibrium toward DMT formation and achieving yields exceeding 90 % [37,38]. Despite these advantages, the depolymerisation of PET under catalyst-free conditions, using DMC as the primary reaction solvent, has not been reported. To address this gap, we characterised the thermolytic behaviour of PET using thermogravimetric analysis (TGA) and pyro-GC/MS. Additionally, pseudo-catalytic depolymerisation was investigated using methanol and/or DMC at various reaction temperatures.

2. Materials and methods

2.1. Sample and chemical reagents

Commercial transparent PET water bottle waste from a single brand was collected from Korea. The collected PET waste was washed, dried, and cut into 3–5 mm flakes. Prior to further analyses, the chemical composition and structure of the collected samples were identified. The elemental composition was analysed using an elemental analyser (EA3000, EuroVector, Italy). The specific functional groups in the PET waste were confirmed using Fourier transform infrared (FT-IR) spectroscopy (Cary 630, Agilent, USA). PET (product#: 429252), DMC (product#: 517127, ≥99 %), DMT (product#: 185124, ≥99 %), EG (product#: 102466, ≥99 %), methanol (product#: 1.06018, ≥99.9 %), and silica (particle size: 60–100 mesh, pore size: 60 Å) were obtained from Sigma-Aldrich (USA). Chloroform (product# 67-66-3, 99.5 % purity) was obtained from Daejung Chemicals (Korea).

2.2. Characterisation of thermal degradation behaviour of PET waste

The thermal degradation behaviour of the PET waste was observed using a TGA unit (STA 449 Jupiter, Netzsch, Germany). The TGA test was performed over a temperature range between 40 and 900 °C at a constant heating rate of 10 °C min^{-1} under an inert (N_2) atmosphere. A sample weighing 10.00 ± 0.01 mg was placed in an alumina crucible. To minimise errors in mass change owing to buoyancy effects at high temperatures, a blank run was conducted prior. In addition, the chemical constituents produced during the PET thermolysis were identified using a pyrolysis-gas chromatography/mass spectrometer (py-GC/MS) equipped with a pyrolyser (Rx-3050TG, Frontier Lab, Japan). A sample weighing 0.1 ± 0.1 mg was inserted into a pyrolyser and subjected to thermolysis under specific temperature intervals (340–360 °C, 360–380 °C, 380–400 °C, 400–420 °C, 440–460 °C, and 460–480 °C) at a heating rate of 10 °C min^{-1}. The gaseous products generated from the pyrolyser were analysed using a GC/MS (8890/5977B, Agilent, USA) equipped with a UA-5MS/HT (30 m × 0.25 mm × 0.25 μm) column.

2.3. Pseudo-transesterification process

A stainless-steel bulkhead (No. 2507-400-61, Swagelok, USA) was used for the bath-type tubular reactor. The internal reactor volume was approximately 1 mL, and the system was suitable for operation at temperature up to 500 °C. First, 160 mg of silica was loaded into the reactor, followed by 10.00 ± 0.01 mg of PET waste and 2.8 mmol of solvent (methanol, DMC, or a mixture). The exact solvent volumes used for each

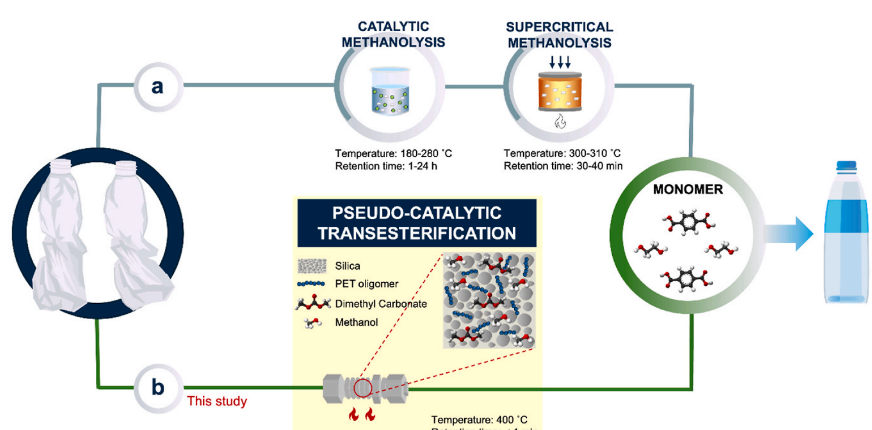

Fig. 1. Pseudo-catalytic transesterification process of PET waste. (a, b) Schematic representation of the PET waste depolymerisation conditions. (a) Conventional catalytic and supercritical methanolysis processes reported in literature. (b) Pseudo-catalytic transesterification of the PET waste.

methanol and DMC composition are reported in Table S1. An additional 160 mg of silica was added. The opposite side of the bulkhead was sealed, and the assembly was placed in a tubular furnace (FT-830, Daihan Scientific) and heated to the target temperature within 1 min. The reaction proceeded under autogenous pressure starting from 1 atm. The internal temperature of the reactor was measured using a K-type thermocouple (OMEGA WI, Madison, USA). No inert gas purging was applied; however, the rector was nearly fully packed with silica, the remaining headspace was minimal, substantially restricting any residual oxygen within the system during heating. After completion of the reaction, the sealed bulkhead was rapidly cooled to 25 °C by forced-air cooling using a cooling fan. A sample diluted 30-fold in chloroform was injected into a GC/MS (5977 B, Agilent, USA) and GC/flame ionisation detector (GC/FID) (8890, Agilent, USA) equipped with a DB-Wax column (30 m × 0.25 mm × 0.25 μm). Quantification of DMC, EG, and EC was conducted by GC/FID using external calibration curves generated from standards. Standard solutions with known concentrations were analysed to construct peak area calibration curves, and the mass of each component was calculated based on the corresponding calibration equation. The DMC, EG, and EC yields from the PET waste were calculated using Eq. (1). To ensure reproducibility, each experiment based on the given test matrix was performed in triplicate.

$$\text{Yield } (\%) = \frac{\textit{Actural weight of DMC, EG, or EC } [g]}{\textit{The initial weight of PET waste sample } [g]} \times 100. \tag{1}$$

2.4. Economic and environmental analysis

The economic and environmental assessments were conducted to evaluate the feasibility of the pseudo-catalytic transesterification process for PET recycling. All calculations were normalized to the depolymerisation of 1 tonne of PET waste at 400 °C using a 1:1 methanol:DMC mixture and a reaction time of 1 min, based on experimentally measured monomer yields (93.1 % DMT and 34.9 % EG). Only raw material and heating energy requirements were considered, and labor, capital cost, and amortization were excluded, providing a conservation process-level estimate.

Raw-material cost was calculated using the following Eq. (2):

$$C_{raw} = \sum_i (m_i \cdot P_i) \tag{2}$$

Where m_i represents the mass of raw material i (methanol, DMC, silica) required per tonne of PET, and P_i is the corresponding unit price ($ per tonne). The market prices of fossil-based PET, DMC, methanol, and silica were assumed to be 1162 [39], 888 [40], 326 [41], and 537 [66] $ per tonne, respectively.

The thermal energy required to heat 1 tonne of PET temperature from 25 to 400 °C was calculated using Eq. (3) assuming a specific heat capacity of 1.0 $kJkg^{-1}K^{-1}$ and ideal furnace efficiency ($\eta = 1$), yielding 104.2 kWh per tonne.

$$Q = mc\Delta T \tag{3}$$

Based on an industrial electricity price of $0.068 per kWh^{-1} [42], the electrical-energy demand corresponds to 7.08 $ per tonne of PET processed. Heat loss, latent heats, heat losses, reactor/solvent heating, and furnace inefficiencies were not included, making this a conservative lower-bound estimate.

To estimate recycled PET (rPET) production, DMT and EG were assumed to react at a 0.5 M ratio to form BHET with 95 % conversion. efficiency [43], followed by complete polymerisation. Under these assumptions, the pseudo-catalytic transesterification process achieves a 68 % PET to rPET conversion rate.

Environmental benefits were quantified by avoided PET disposal emissions. Literature emission factors for PET incineration and landfill were applied (6.2 and 4.3 tonnes of CO_2 are released per tonne of PET, respectively [44]). Disposal costs were taken 605 $ per tonne for incineration [45] and 61 $ per tonne for landfill [46], respectively.

3. Results and discussion

3.1. Characterisation of PET waste

Elemental analysis was conducted to accurately identify the collected PET bottle waste (Fig. 2(a)). The samples consisted of C (63.86 wt%), H (4.31 wt%), O (31.83 wt%), and N (0.22 wt%). Because PET typically contains 62.20–64.10 wt % of C, 3.70–4.21 wt% of H, 33.29–34.20 wt% of O, and trace amounts of N (0–0.14 wt%) and S (0–0.32 wt%) [47–50], the chemical composition of the sample was consistent with that of PET. Fourier transform infrared (FT-IR) spectroscopy (Cary 630, Agilent, USA) confirmed that the spectrum of the sample matched that of PET (Fig. 2(a)).

To depolymerise PET waste through pseudo-catalytic transesterification, which is a thermally induced reaction, its thermal degradation behaviour must be investigated. Accordingly, TGA was conducted in a dynamic temperature regime in an inert N_2 atmosphere. The mass changes and DTG peaks of PET waste are shown in Fig. 2(c). PET waste undergoes single-stage degradation between 354 and 485 °C, with a maximum DTG peak at 435 °C. Seventy-eight wt.% of the PET waste was thermally decomposed within this temperature range. This temperature range corresponds to the devolatilisation zone, where thermal degradation of the PET polymeric backbone occurs. Above 485 °C, a slow mass change of PET waste was observed, which could be attributed to carbonisation (char formation). During this phase, the thermolysis of PET initiates cross-linking reactions among the degradation products, leading to the formation of polyaromatics, which are considered precursors for char formation [51]. At 900 °C, the residual mass of the PET waste was measured at 15.53 wt%. These cross-linking reactions are considered undesired side reactions because they lead to the formation of thermally stable char, which hinders complete depolymerisation and reduces the overall efficiency of the PET recovery processes.

To gain deeper insight into the thermal degradation behaviour of PET waste within the devolatilisation zone (354–485 °C), the volatile chemical compounds evolved at specific temperature intervals, namely, 340–360 °C, 360–380 °C, 380–400 °C, 400–420 °C, 440–460 °C, and 460–480 °C, were analysed using pyrolysis-gas chromatography/mass spectrometer (py-GC/MS), as shown in Fig. 3(a). The potential thermal degradation pathways based on py-GC/MS analysis are shown in Fig. 3 (b).

As illustrated in Fig. 3(a), acetaldehyde is identified as the initial degradation product of PET waste in the temperature range between 340 and 380 °C. This observation indicates that the thermal degradation of the PET polymeric backbone begins within this temperature range, which corresponds to the initial temperature of the mass change observed in Fig. 2(c). Previous studies confirmed that acetaldehyde is the first gaseous product released during the thermolysis of PET [52,53]. In the higher temperature range between 380 and 400 °C, additional compounds, such as carbon dioxide, benzene, vinyl benzoate, benzoic acid, propyl benzoate, 2-(methoxycarbonyl)benzoic acid, 4-(methoxycarbonyl)benzoic acid, benzoic anhydride, 2-(benzoyloxy)ethyl vinyl terephthalate, ethyl (4-isopropylphenyl) phthalate, and 3,6,10,13-tetraoxa-1,8 (1,4)-dibenzenacyclotetradecaphane-2,7,9,14-tetraone were identified. These compounds most probably form via the random cleavage of alkyl bonds in cyclic oligomers, a process driven by β-CH hydrogen transfer [54]. This process is facilitated by the low enthalpy of the C-O bond and the presence of hydrogen at the α-position on the carbon atoms [55]; [56]. As the temperature increased to 400–420 °C, the peak intensities of all the chemicals increased, indicating accelerated degradation of the PET polymeric backbone. However, at temperatures exceeding 420 °C, the peak intensities decreased, suggesting that secondary cracking reactions dominated this stage. During this phase, high-molecular-weight aromatic compounds and esters undergo further

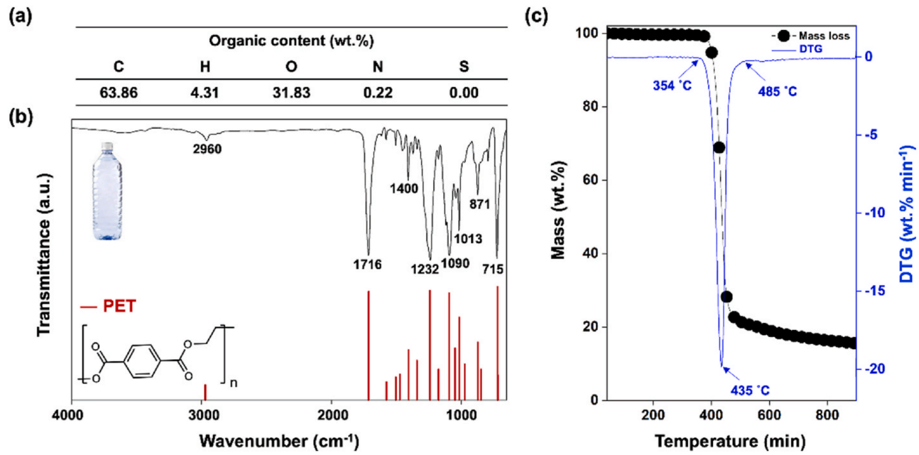

Fig. 2. Characterisation of PET waste. (a) Organic content of sample showing elemental composition (wt.%). (b) FT-IR) spectra with reference wavelengths for PET characteristic peaks (red bars). (c) Thermogravimetric analysis (TGA) showing mass changes and derivative thermogravimetric (DTGs) curves under inert conditions.

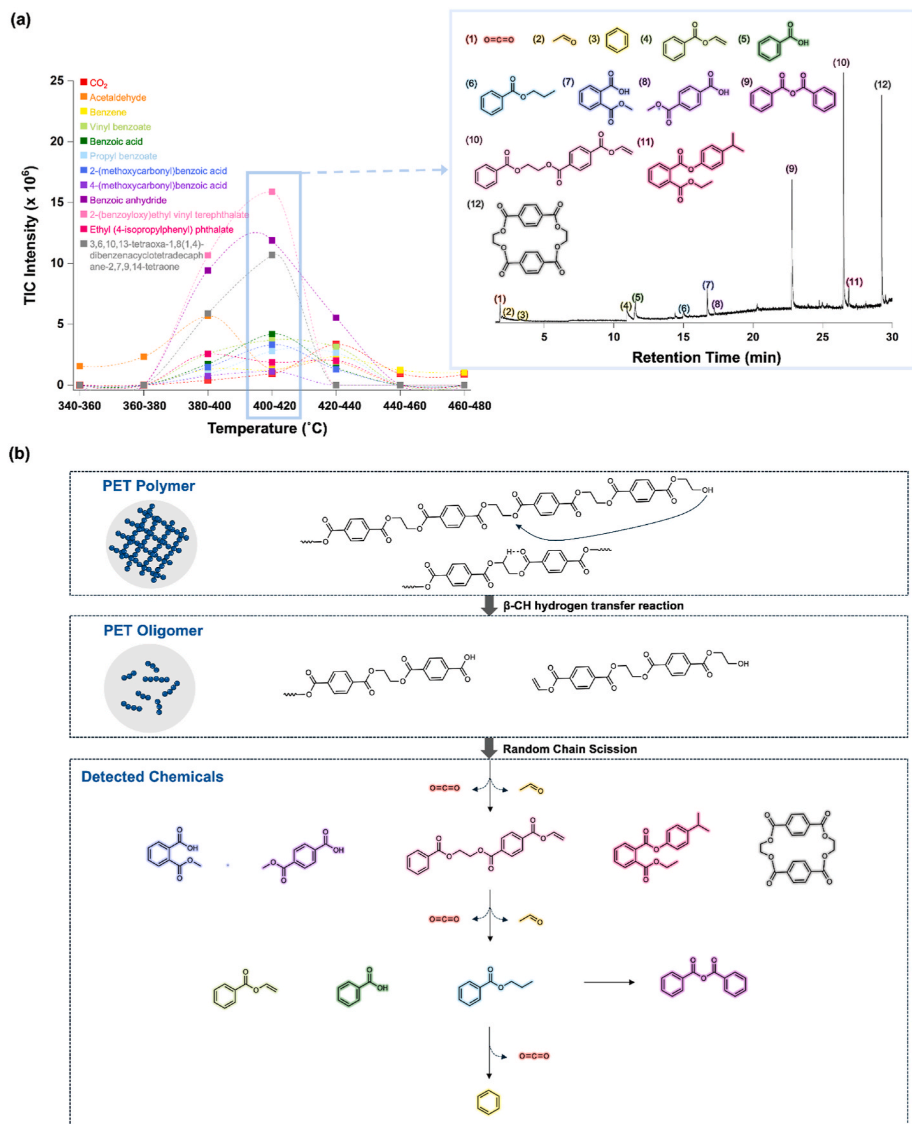

Fig. 3. py-GC/MS analysis and proposed reaction pathways of PET waste. (a) Py-GC/MS analysis of PET waste at different temperature intervals (340–360 °C, 360–380 °C, 380–400 °C, 400–420 °C, 440–460 °C, and 460–480 °C), showing the thermal degradation behaviour and detected products. (b) Proposed reaction pathways for PET decomposition, illustrating β-CH hydrogen transfer reactions, oligomer formation, and subsequent random chain scission leading to various detected chemicals.

degradation through random scission of polymeric chains, resulting in the formation of lower-molecular-weight hydrocarbons and volatile fragments. As a result, most volatile compounds decompose into gaseous by-products, such as methane, ethylene, and carbon monoxide [57]. Notably, between 460 and 480 °C, only carbon dioxide and benzene were detected. This observation suggests that decarboxylation, the elimination of carbon dioxide from the terminal carboxyl groups in the PET structure, plays a dominant role at this stage. This reaction stabilises the PET backbone by forming phenyl terminals such as benzene [58]. This occurrence indicates that PET thermal degradation becomes less effective for monomer recovery above 420 °C owing to the increased formation of gaseous by-products.

3.2. Pseudo-catalytic transesterification of PET waste with methanol

Depolymerisation of PET begins with the breakdown of PET chains into oligomers [59,60]. The thermal energy supplied by the external heating source initiates the depolymerisation process. Given that the transesterification reaction (*i.e.*, methanolysis) typically occurs at 200 °C in the presence of a catalyst [60–62], catalyst-free monomer conversion requires a temperature exceeding 200 °C. However, above this temperature, the interaction time between the two-phase reactants (*i.e.,* molten PET and gaseous methanol) is considerably reduced, hindering efficient monomer conversion. To overcome this challenge, this study explored the use of porous materials (*i.e.,* silica) to prolong the interaction time between the reactants. The incorporation of porous materials is expected to increase the collision frequency between molten PET and gaseous methanol within the pores, thereby inducing a catalytic-like effect that lowers the activation energy of the transesterification reaction. Since methanolysis proceeds through initial thermally induced chain scission to form oligomers, thermal decomposition experiments were performed as a baseline to distinguish purely thermal decomposition from methanol-assisted depolymerisation. This approach confirms that the role of silica is associated with enhancing PET-methanol contact rather than modifying the intrinsic thermal degradation behavior of PET. Before conducting the pseudo-catalytic transesterification reaction, we investigated the effect of silica on the thermal degradation behaviour of PET waste. The degradation profile of the PET waste in the presence of silica was consistent with the patterns obtained from the py-GC/MS analyses (Fig. S1). The lack of significant changes in the decomposition products or peak intensities indicates that silica had a negligible impact on the thermal degradation kinetics of PET waste.

The pseudo-catalytic transesterification reaction between PET and methanol was conducted for 1 min within a temperature range of 280 °C (considering the melting point of PET) to 420 °C. The upper temperature limit of 420 °C was selected because it corresponds to the onset of secondary cracking, as shown in Fig. 3(a). The yields of DMT and EG from the pseudo-catalytic transesterification reaction are plotted in Fig. 4 as a function of reaction temperature. At temperatures below 340 °C, the yields of DMT and EG remained marginal, likely because of insufficient thermal energy to efficiently break the ester linkages in the PET polymeric backbone. Above 340 °C, the yields increased rapidly in proportion to the reaction temperature, as shown by the enhanced decomposition rates and volatile product formation observed in Fig. 2(c). Transesterification of PET with methanol is an endothermic and reversible process. An increase in temperature provides sufficient activation energy to drive the reaction forward and shifts the equilibrium towards monomer formation. Notably, at 400 °C, the DMT and EG yields reached 75.0 and 17.0 %, respectively, within 1 min. Accomplishing this result within 1 min highlights the economic feasibility of the pseudo-catalytic approach. Accelerated reaction kinetics minimise the overall energy consumption, making the process more cost-effective and scalable for industrial applications. However, at 420 °C, the yields of DMT and EG decreased, which is consistent with the secondary thermal cracking observed in the pyrolysis experiments (Fig. 3). At this temperature, monomeric species and low-molecular-weight intermediates undergo further fragmentation into gaseous products, resulting in a reduction in recoverable monomer yield.

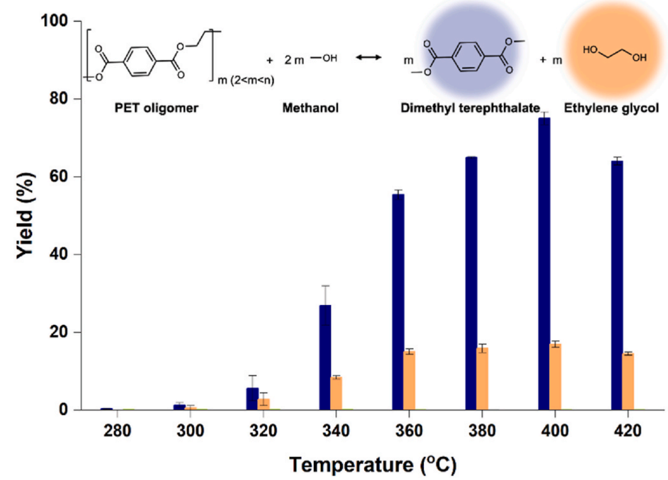

Fig. 4. Yields of DMT and EG from pseudo-catalytic transesterification of PET waste with methanol. Yields of DMT and EG at different reaction temperatures illustrating the effect of temperature on the pseudo-catalytic transesterification process. Error bars indicate the standard deviation from three independent experiments.

3.3. Pseudo-catalytic transesterification of PET waste with DMC

To enhance monomer recovery efficiency, DMC was utilised as an alternative solvent for the pseudo-catalytic transesterification of PET waste. Hypothetically, the electrophilic carbonyl carbon of DMC could facilitate ester bond cleavage in PET [35], thereby promoting depolymerisation efficiency. Under thermal conditions, DMC is also expected to undergo *in-situ* decomposition into methanol, potentially driving the reaction equilibrium towards monomer formation. To verify this hypothesis, the pseudo-catalytic transesterification of PET waste was conducted using methanol-DMC mixtures with various molar ratios as a function of the reaction temperature.

Fig. 5(a–c) show the results of the pseudo-catalytic transesterification using methanol:DMC molar ratios of 2.5:1, 1:1, and 1:2.5, respectively, whereas Fig. 5(d) shows the reaction performed using DMC alone (0:1). The methanol-DMC mixture yielded a higher monomer recovery than that of methanol alone (Fig. 4). For example, the DMT yield reached 75.0 % at 400 °C with methanol, whereas it increased to 88.6 % at a lower temperature of 380 °C with a 2.5:1 methanol/DMC mixture. Similarly, the EG yield was 17.0 % at 400 °C with methanol, whereas it increased to 25.2 % at 380 °C with the 2.5:1 methanol:DMC mixture. Notably, higher yields of DMT and EG were obtained when a 1:1 M ratio of methanol/DMC was used. The enhanced reaction kinetics, upon the addition of DMC, can be attributed to its electrophilic carbonyl carbon, which facilitates nucleophilic attack by the ethylene glycol segment of the PET oligomers [35]. However, as the proportion of DMC in the mixture increased beyond 1:1, the recovery rate of EG decreased, whereas the yield of EC increased. When DMC alone was used, the EC yield increased further, reaching 10.8 % at 360 °C (Fig. 5(d)). This suggests that as the proportion of DMC increased, EG was increasingly converted into the more thermodynamically stable EC, thereby reducing EG recovery. However, considering the re-polymerisation process for producing recycled PET (rPET), the production of DMT and EG is crucial. Note that all product yields are normalized to the initial mass of PET. Therefore, values exceeding 100 % arise because DMC contributes additional methyl/carbonyl units and in-situ-generated methanol further participates in transesterification, resulting in product mass

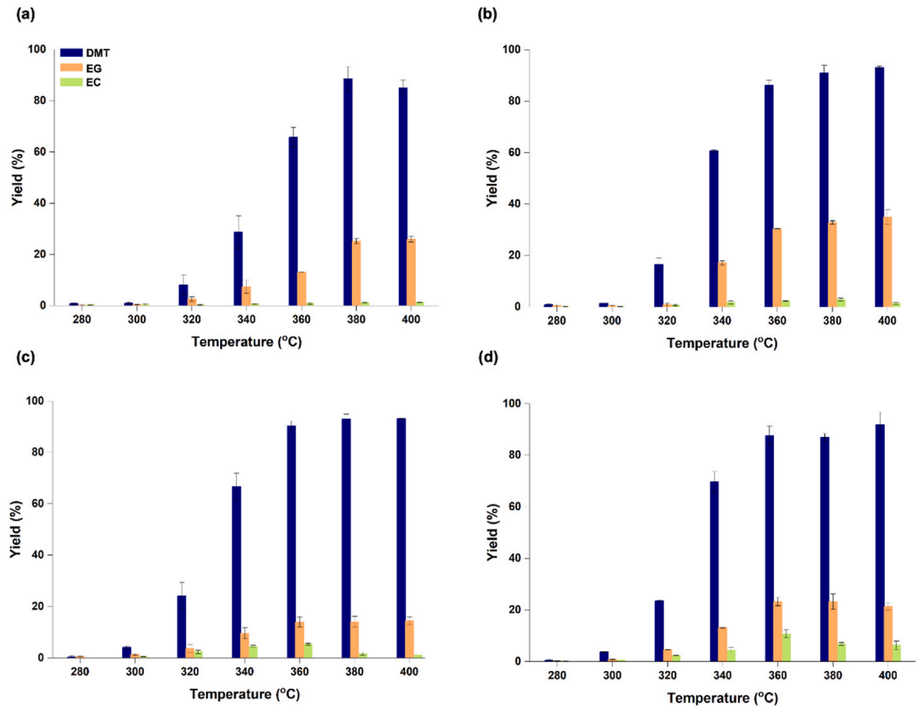

Fig. 5. Yields of DMT, EG, and EC from pseudo-catalytic transesterification of PET waste. (a) Reaction with a 2.5:1 M ratio of methanol to DMC. (b) Reaction with 1:1 M ratio of methanol to DMC. (c) Reaction with a 1:2.5 M ratio of methanol and DMC. (d) Reaction with only DMC as the reactant.

greater than that originating from PET alone. In practice, DMT and EG undergo transesterification at a 1:2 M ratio to produce *bis*(2-hydroxyethyl) terephthalate (BHET), which is then processed into PET through a polycondensation reaction [43]. Although DMC can depolymerise PET waste more effectively than methanol, a 1:1 methanol:DMC mixture is considered the optimal solvent for simultaneously achieving high DMT and EG recovery efficiencies.

Based on these observations, a possible pseudo-catalytic transesterification mechanism for PET waste using DMC is shown in Fig. 6. DMT and EC are generated by replacing the hydroxyl groups in the EG segment of the PET oligomers with methyl groups from DMC. Additionally, DMT and EG can be formed via transesterification, which is facilitated by the methanol produced from the thermal cracking of DMC. The EG formed in this reaction can further react with DMC, yielding a more stable EC and additional methanol [37]. Subsequently, the methanol formed in situ can further participate in transesterification with PET oligomers, leading to the formation of DMT and EG. Although these reactions occur simultaneously, the higher yield of EG relative to that of EC when only DMC is used (Fig. 5(d)) indicates that the transesterification reaction involving methanol remains the predominant reaction pathway. The competition between EG recovery and EC formation depended on the methanol:DMC ratio, which influenced the overall depolymerisation efficiency (see Fig. 5).

In summary, DMC accelerates the reaction kinetics for the depolymerisation of PET waste by facilitating nucleophilic attack owing to its electrophilic nature. However, the formation of EC as a byproduct

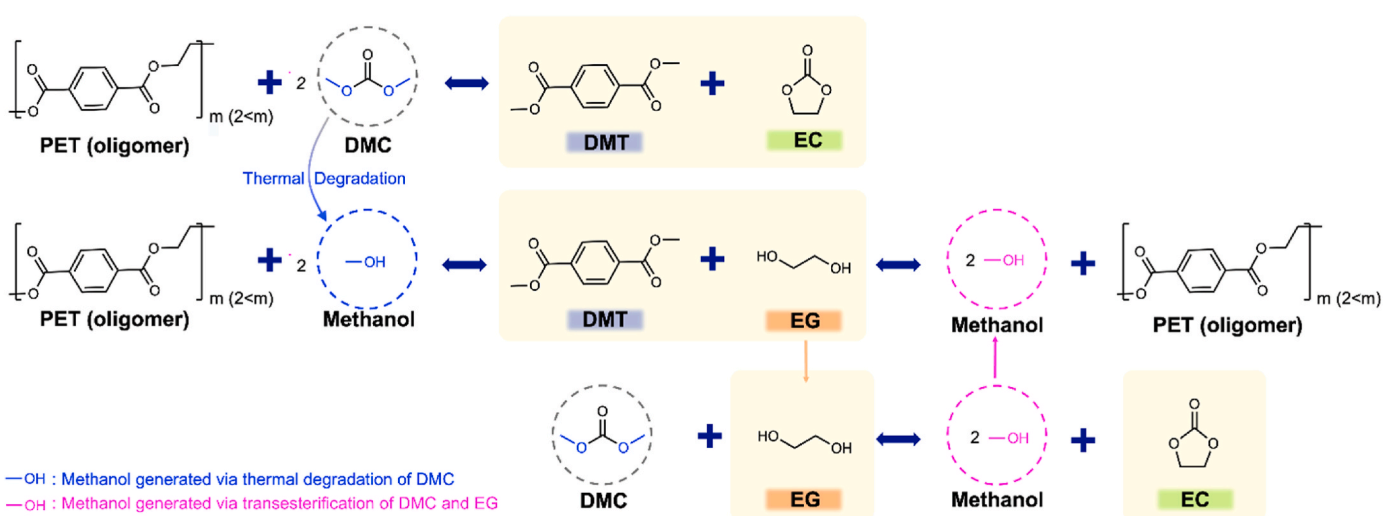

Fig. 6. Pseudo-catalytic transesterification mechanism of PET waste with DMC. Illustration of possible reaction pathways for PET oligomer depolymerisation using DMC. DMC undergoes direct transesterification with a PET oligomer and is thermally decomposed into methanol for further transesterification. DMC can react with EG to produce methanol and EC, whereas the produced methanol can react with the PET oligomers again to produce DMT and EG.

during the transesterification of PET waste with DMC reduced the overall monomer recovery efficiency. Importantly, the reactions involving methanol remained the predominant pathway in the pseudo-catalytic transesterification process with DMC. Moreover, the presence of methanol did not negatively affect the DMC-enhanced reaction kinetics. By using a methanol-DMC mixture as a solvent, EC production can be minimised while increasing EG recovery. Consequently, introducing an optimised 1:1 methanol:DMC mixture could serve as a viable strategy to mitigate excessive byproduct formation by reducing the proportion of carbon diverted into the EC, thereby improving the efficiency of monomer recovery.

3.4. Economic and environmental aspects of pseudo-catalytic transesterification of PET waste with DMC

The objective of this study is to develop an economically viable depolymerisation process for PET waste. Operating conditions, including temperature, pressure, reaction time, and catalyst usage, critically affect process efficiency and are directly linked to operating costs. To assess the process efficiency, the results obtained from the pseudo-catalytic transesterification process using a 1:1 methanol:DMC mixture were compared with those obtained from the conventional methanolysis process, as summarised in Table 1. Conventional methanolysis has achieved a high yield of DMT at relatively lower temperatures (25–200 °C) in the presence of a catalyst, whereas, in the absence of a catalyst, a high yield of DMT has been obtained under supercritical conditions (300 °C, 8.3–14.7 Mpa). Both approaches require prolonged reaction times (0.5–24 h). By contrast, the pseudo-catalytic transesterification process operates at a higher temperature (400 °C) but achieves a significantly reduced reaction time (<1 min) while eliminating the need for a catalyst. To quantitatively compare the process efficiencies, the reaction performance metric (RPM), which is defined as the yield of DMT produced per unit temperature and time, was introduced. To allow direct comparison, all reaction parameters except temperature and reaction time (i.e., solvent-to-PET ratio, reactor volume, heating profile, and charge amount) were held constant. Therefore, RPM was applied as a controlled performance metric within this fixed experimental framework, while other potential influencing factors are reserved for future optimization. Despite the higher temperature required, the pseudo-catalytic transesterification process exhibited the highest RPM among those of the reported methods owing to its ultrafast reaction time. Considering that RPM does not account for the presence of a catalyst, this performance suggests that the pseudo-catalytic transesterification process exhibits a superior performance to those of the other processes.

Economic feasibility was evaluated based on raw material cost and energy consumption (Fig. 7). The economic assessment was conducted at an optimal reaction temperature of 400 °C for the pseudo-catalytic transesterification process using a 1:1 methanol:DMC mixture. Under these conditions, the yields of DMT and EG were 93.1 and 34.9 %, respectively. For the depolymerisation of 1 tonne of PET waste, the process required \$160 of methanol, \$1101 of DMC, and \$1718 of silica. The process also requires 104.17 kWh of electrical energy, resulting in an additional cost of \$7.08, based on an average industrial electricity price of \$0.068 per kWh [42]. To evaluate the economic output, DMT and EG were assumed to react at a molar ratio of 0.5 to form BHET with a conversion efficiency of 95 % [43], followed by its complete polymerisation into rPET under optimal conditions. Based on these assumptions, the pseudo-catalytic transesterification process achieves an rPET conversion rate of 68 % from PET waste, with an estimated economic value of \$4392 per tonne of rPET produced.

Although the production cost of rPET via pseudo-catalytic transesterification (4392 \$/tonne) remains significantly higher than the production and disposal cost [45,46] of virgin PET derived from fossil fuels (1223–1767 \$/tonne) [39], PET recycling plays a crucial role in both economic and environmental sustainability, by reducing the dependence on virgin fossil resources and mitigating the CO_2 emissions associated with PET. PET generates significant CO_2 emissions throughout its life cycle, including plastic production, product manufacturing, waste transportation, and disposal. Notably, approximately 6.2 tonnes of CO_2 are released per tonne of PET waste when incinerated, while landfill disposal results in 4.3 tonnes of CO_2 emissions over its life cycle [44]. Given the short lifespan of PET bottles, their rapid accumulation as waste poses a critical environmental challenge [64]. Virgin PET production alone emit approximately 2.9–3.5 tonnes of CO_2 per tonne of PET [44]. Based on the reported recycling efficiency of 68 % for PET recovery, recycling 1 tonne of PET via the pseudo-catalytic transesterification process can avoid approximately 2 tonnes of CO_2 emissions by displacing virgin PET production. This simplified carbon estimate focuses on avoided production emissions; additional benefits from preventing incineration/landfill-related emissions would further enhance the net CO_2 reduction and will be evaluated in future work. Recognising the importance of PET recycling, the European Union has established mandatory recycled content targets for plastic beverage bottles of 30 % by 2030 and 65 % by 2040 [65]. Such regulations, along with increasing consumer awareness, are expected to drive the demand for rPET, reduce reliance on volatile fossil fuel prices, and improve resource efficiency in the plastic industry.

Although the current production cost of the pseudo-catalytic transesterification process remains high, further advances and process optimization have the potential to significantly reduce the manufacturing expenses, thereby enhancing the economic feasibility of this recycling method. One of the major challenges is the excessive consumption of solvents (methanol and DMC), necessitating optimization to improve efficiency. In addition, the high consumption of silica contributes to increased costs. Therefore, further research is required to evaluate the feasibility of silica reuse. With continuous advances in process technology, the production cost of rPET is expected to decrease, thereby contributing to the realisation of a circular economy.

4. Conclusions

We developed a catalyst-free depolymerisation technology for PET waste using porous materials, referred to as the pseudo-catalytic transesterification process. The incorporation of silica facilitated enhanced reaction kinetics, achieving high DMT and ethylene glycol (EG) yields

Table 1
Comparison of reported PET chemical depolymerisation conditions and yields.

Reagent	Temperature, °C	Time, h	Catalyst	DMT Yield, %	RPM[a], h^{-1} °C^{-1}	Reference
methanol -DMC (1:1)	400	0.02	–	93.1	11.6	This study
methanol	200	0.5	MgO/NaY	91	0.9	[62]
methanol	200	0.5	Calcined sodium silicate	95	1.0	[60]
methanol	200	1.5	Cu/SiO_2	91.4	0.3	[63]
methanol	300	0.67	–	97	0.5	[32]
methanol	300	0.5	–	98	0.7	[31]

[a] RPM calculated by the equation: $\frac{\text{DMT Yield}}{\text{Time} \times \text{Temperature}}$

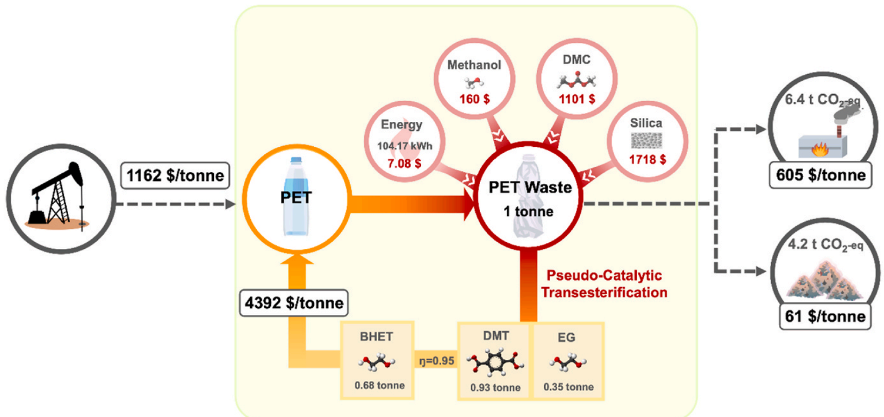

Fig. 7. Economic analysis of PET waste recycling via pseudo-catalytic transesterification. The economic value was calculated at the optimal yield temperature of 400 °C for the pseudo-catalytic transesterification process using a 1:1 methanol:DMC mixture. The yield of BHET was determined under the assumption of a DMT:EG mole ratio of 0.5, and conversion efficiency of 95 % [43]. The produced BHET was assumed to have undergone complete polymerisation into PET without loss. The market prices of fossil-based PET, DMC, methanol, and silica were 1162 [39], 888 [40], 326 [41], and 537 [66]$ per tonne, respectively. The disposal cost of plastic waste by incineration and landfill is 605 $/tonne [45] and 61 $/tonne [46], respectively.

(75.0 and 17.0 %, respectively) within 1 min at 400 °C. To further improve the monomer recovery, dimethyl carbonate (DMC) was introduced as a (co)solvent to replace methanol. The results demonstrated that a methanol:DMC mixture outperformed methanol alone, with a methanol:DMC molar ratio of 1:1 yielding the highest DMT and EG recovery. However, an excessive proportion of DMC promoted ethylene carbonate (EC) formation, which led to a reduction in the EG yield. Therefore, optimising the methanol-DMC ratio (1:1) is crucial for minimising EC production while maximising the monomer recovery efficiency. As a result of this optimised process, the pseudo-catalytic transesterification of PET waste achieved a recycled PET (rPET) conversion rate of 68 % with an estimated economic value of $4392 per tonne of rPET produced. Despite the higher production cost compared with that of fossil-based PET, this process offers substantial environmental benefits by reducing the production of approximately 2 tonnes of CO_2 per tonne of recycled PET waste. As sustainability has become a global priority, regulatory policies and increasing consumer awareness are expected to drive the rPET demand and promote market adoption. Future optimization efforts, such as solvent minimization, silica reuse, and time-resolved reaction profiling under varied heating conditions, will further improve economic and environmental performance. These advancements will enhance the industrial viability of pseudo-catalytic transesterification for PET recycling and contribute to the realisation of a circular economy.

CRediT authorship contribution statement

Dohee Kwon: Writing – review & editing, Writing – original draft, Visualization, Investigation, Conceptualization. **Doyeon Lee:** Writing – review & editing, Writing – original draft, Validation, Resources, Visualization. **Jechan Lee:** Writing – review & editing, Validation, Methodology. **Jee Young Kim:** Writing – review & editing, Visualization, Validation. **Eilhann E. Kwon:** Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.polymertesting.2025.109051.

Data availability

Data will be made available on request.

References

[1] J. Cao, H. Liang, J. Yang, Z. Zhu, J. Deng, X. Li, M. Elimelech, X. Lu, Depolymerization mechanisms and closed-loop assessment in polyester waste recycling, Nat. Commun. 15 (1) (2024) 6266.
[2] Y. Ke, C. Long, Z. Qi, Crystallization, properties, and crystal and nanoscale morphology of PET–clay nanocomposites, J. Appl. Polym. Sci. 71 (7) (1999) 1139–1146.
[3] T. Muringayil Joseph, S. Azat, Z. Ahmadi, O. Moini Jazani, A. Esmaeili, E. Kianfar, J. Haponiuk, S. Thomas, Polyethylene terephthalate (PET) recycling: a review, Case Stud. Chem. Environ. Eng. 9 (2024) 100673.
[4] E. Luna, I. Olazabal, M. Roosen, A. Müller, C. Jehanno, M. Ximenis, S. de Meester, H. Sardon, Towards a better understanding of the cosolvent effect on the low-temperature glycolysis of Polyethylene Terephthalate (PET), Chem. Eng. J. 482 (2024) 148861.
[5] X. Yuan, S. Li, S. Jeon, S. Deng, L. Zhao, K.B. Lee, Valorization of waste polyethylene terephthalate plastic into N-doped microporous carbon for CO_2 capture through a one-pot synthesis, J. Hazard Mater. 399 (2020) 123010.
[6] Z.-H. Qin, J.-H. Mou, C.Y.H. Chao, S.S. Chopra, W. Daoud, S.-y. Leu, Z. Ning, C. Y. Tso, C.K. Chan, S. Tang, Z.J. Hathi, M.A. Haque, X. Wang, C.S.K. Lin, Biotechnology of plastic waste degradation, recycling, and valorization: current advances and future perspectives, ChemSusChem 14 (19) (2021) 4103–4114.
[7] M.W. Ryberg, M.Z. Hauschild, F. Wang, S. Averous-Monnery, A. Laurent, Global environmental losses of plastics across their value chains, Resour. Conserv. Recycl. 151 (2019) 104459.
[8] C. Bharadwaj, R. Purbey, D. Bora, P. Chetia, U. Maheswari R, R. Duarah, K. Dutta, E.R. Sadiku, K. Varaprasad, J. Jayaramudu, A review on sustainable PET recycling: strategies and trends, Mater. Today Sustain. 27 (2024) 100936.
[9] Z. Jiang, D. Yan, J. Xin, F. Li, M. Guo, Q. Zhou, J. Xu, Y. Hu, X. Lu, Poly(ionic liquid)s as efficient and recyclable catalysts for methanolysis of PET, Polym. Degrad. Stabil. 199 (2022) 109905.
[10] A. McNeeley, Y.A. Liu, Assessment of PET depolymerization processes for circular economy. 2. Process design options and process modeling evaluation for

methanolysis, glycolysis, and hydrolysis, Ind. Eng. Chem. Res. 63 (8) (2024) 3400–3424.
[11] H. Wu, S. Lv, Y. He, J.-P. Qu, The study of the thermomechanical degradation and mechanical properties of PET recycled by industrial-scale elongational processing, Polym. Test. 77 (2019) 105882.
[12] I. Amundarain, S. López-Montenegro, L. Fulgencio-Medrano, J. Leivar, A. Iruskieta, A. Asueta, R. Miguel-Fernández, S. Arnaiz, B. Pereda-Ayo, Improving the sustainability of catalytic glycolysis of complex PET waste through bio-solvolysis, Polymers 16 (1) (2024) 142.
[13] J. Cao, X. Qiu, F. Zhang, S. Fu, Circular economy and chemical conversion for polyester wastes, ChemSusChem e202402100.
[14] G.W. Coates, Y.D.Y.L. Getzler, Chemical recycling to monomer for an ideal, circular polymer economy, Nat. Rev. Mater. 5 (7) (2020) 501–516.
[15] A.B. Raheem, Z.Z. Noor, A. Hassan, M.K. Abd Hamid, S.A. Samsudin, A.H. Sabeen, Current developments in chemical recycling of post-consumer polyethylene terephthalate wastes for new materials production: a review, J. Clean. Prod. 225 (2019) 1052–1064.
[16] A. McNeeley, Y.A. Liu, Assessment of PET depolymerization processes for circular economy. 1. Thermodynamics, chemistry, purification, and process design, Ind. Eng. Chem. Res. 63 (8) (2024) 3355–3399.
[17] V. Sinha, M.R. Patel, J.V. Patel, Pet waste management by chemical recycling: a review, J. Polym. Environ. 18 (1) (2010) 8–25.
[18] D.D. Pham, J. Cho, Low-energy catalytic methanolysis of poly (ethyleneterephthalate), Green Chem. 23 (1) (2021) 511–525.
[19] G.P. Karayannidis, A.P. Chatziavgoustis, D.S. Achilias, Poly(ethylene terephthalate) recycling and recovery of pure terephthalic acid by alkaline hydrolysis, Adv. Polym. Technol. 21 (4) (2002) 250–259.
[20] Z.T. Laldinpuii, V. Khiangte, S. Lalhmangaihzuala, C. Lalmuanpuia, Z. Pachuau, C. Lalhriatpuia, K. Vanlaldinpuia, Methanolysis of PET waste using heterogeneous catalyst of bio-waste origin, J. Polym. Environ. 30 (4) (2022) 1600–1614.
[21] D. Paszun, T. Spychaj, Chemical recycling of poly(ethylene terephthalate), Ind. Eng. Chem. Res. 36 (4) (1997) 1373–1383.
[22] F. Cao, L. Wang, R. Zheng, L. Guo, Y. Chen, X. Qian, Research and progress of chemical depolymerization of waste PET and high-value application of its depolymerization products, RSC Adv. 12 (49) (2022) 31564–31576.
[23] J. Li, D. Yan, X. Cheng, C. Rong, J. Feng, X. Feng, J. Xin, Q. Zhou, Y. Li, J. Xu, X. Lu, Efficient methanolysis of PET catalyzed by nonmetallic deep eutectic solvents, Ind. Eng. Chem. Res. 63 (28) (2024) 12373–12384.
[24] S. Mishra, A.S. Goje, Kinetic and thermodynamic study of methanolysis of poly (ethylene terephthalate) waste powder, Polym. Int. 52 (3) (2003) 337–342.
[25] P. Anggo Krisbiantoro, Y.-W. Chiao, W. Liao, J.-P. Sun, D. Tsutsumi, H. Yamamoto, Y. Kamiya, K. C-W Wu, Catalytic glycolysis of polyethylene terephthalate (PET) by solvent-free mechanochemically synthesized MFe_2O_4 (M = Co, Ni, Cu and Zn) spinel, Chem. Eng. J. 450 (2022) 137926.
[26] M. Imran, D.H. Kim, W.A. Al-Masry, A. Mahmood, A. Hassan, S. Haider, S. M. Ramay, Manganese-, cobalt-, and zinc-based mixed-oxide spinels as novel catalysts for the chemical recycling of poly(ethylene terephthalate) via glycolysis, Polym. Degrad. Stabil. 98 (4) (2013) 904–915.
[27] R.-X. Yang, Y.-T. Bieh, C.H. Chen, C.-Y. Hsu, Y. Kato, H. Yamamoto, C.-K. Tsung, K. C.W. Wu, Heterogeneous metal azolate Framework-6 (MAF-6) catalysts with high zinc density for enhanced polyethylene terephthalate (PET) conversion, ACS Sustain. Chem. Eng. 9 (19) (2021) 6541–6550.
[28] F. Liu, X. Cui, S. Yu, Z. Li, X. Ge, Hydrolysis reaction of poly(ethylene terephthalate) using ionic liquids as solvent and catalyst, J. Appl. Polym. Sci. 114 (6) (2009) 3561–3565.
[29] F. Chen, G. Wang, W. Li, F. Yang, Glycolysis of poly(ethylene terephthalate) over Mg–Al mixed oxides catalysts derived from hydrotalcites, Ind. Eng. Chem. Res. 52 (2) (2013) 565–571.
[30] M. Ma, S. Wang, Y. Liu, H. Yu, S. Yu, C. Ji, H. Li, G. Nie, S. Liu, Insights into the depolymerization of polyethylene terephthalate in methanol, J. Appl. Polym. Sci. 139 (35) (2022) e52814.
[31] M. Genta, T. Iwaya, M. Sasaki, M. Goto, T. Hirose, Depolymerization mechanism of poly(ethylene terephthalate) in supercritical methanol, Ind. Eng. Chem. Res. 44 (11) (2005) 3894–3900.
[32] B.-K. Kim, G.-C. Hwang, S.-Y. Bae, S.-C. Yi, H. Kumazawa, Depolymerization of polyethyleneterephthalate in supercritical methanol, J. Appl. Polym. Sci. 81 (9) (2001) 2102–2108.
[33] J.-M. Jung, J.-I. Oh, D. Kwon, Y.-K. Park, M. Zhang, J. Lee, E.E. Kwon, Synthesis of fatty acid methyl esters via non-catalytic transesterification of avocado oil with dimethyl carbonate, Energy Convers. Manag. 195 (2019) 1–6.
[34] E.E. Kwon, S. Kim, Y.J. Jeon, H. Yi, Biodiesel production from sewage sludge: new paradigm for mining energy from municipal hazardous material, Environ. Sci. Technol. 46 (18) (2012) 10222–10228.
[35] G. Fiorani, A. Perosa, M. Selva, Dimethyl carbonate: a versatile reagent for a sustainable valorization of renewables, Green Chem. 20 (2) (2018) 288–322.
[36] D. Fabbri, V. Bevoni, M. Notari, F. Rivetti, Properties of a potential biofuel obtained from soybean oil by transmethylation with dimethyl carbonate, Fuel 86 (5) (2007) 690–697.
[37] S. Tanaka, M. Koga, T. Kuragano, A. Ogawa, H. Ogiwara, K. Sato, Y. Nakajima, Depolymerization of polyester fibers with dimethyl carbonate-aided methanolysis, ACS Mater. Au 4 (3) (2024) 335–345.
[38] S. Tanaka, J. Sato, Y. Nakajima, Capturing ethylene glycol with dimethyl carbonate towards depolymerisation of polyethylene terephthalate at ambient temperature, Green Chem. 23 (23) (2021) 9412–9416.
[39] (ChemIntel360), A.R., Market Value of Polyethylene Terephthalate Worldwide from 2015 to 2022, with a Forecast for 2023 to 2030 (In Billion U.S. Dollars), Statista, 2023.
[40] V. Kontou, D. Grimekis, K. Braimakis, S. Karellas, Techno-economic assessment of dimethyl carbonate production based on carbon capture and utilization and power-to-fuel technology, Renew. Sustain. Energy Rev. 157 (2022) 112006.
[41] W.T. Lang, S.A. Mehta, M.M. Thomas, D. Openshaw, E. Westgate, G. Bagnato, Chemical recycling of polyethylene terephthalate, an industrial and sustainable opportunity for Northwest of England, J. Environ. Chem. Eng. 11 (5) (2023) 110585.
[42] S. LeBlanc, S.K. Yee, M.L. Scullin, C. Dames, K.E. Goodson, Material and manufacturing cost considerations for thermoelectrics, Renew. Sustain. Energy Rev. 32 (2014) 313–327.
[43] M. Di Serio, R. Tesser, A. Ferrara, E. Santacesaria, Heterogeneous basic catalysts for the transesterification and the polycondensation reactions in PET production from DMT, J. Mol. Catal. Chem. 212 (1) (2004) 251–257.
[44] I. Vollmer, M.J. Jenks, M.C. Roelands, R.J. White, T. van Harmelen, P. de Wild, G. P. van der Laan, F. Meirer, J.T. Keurentjes, B.M. Weckhuysen, Beyond mechanical recycling: giving new life to plastic waste, Angew. Chem., Int. Ed. 59 (36) (2020) 15402.
[45] R.H.J.M. Gradus, P.H.L. Nillesen, E. Dijkgraaf, R.J. van Koppen, A cost-effectiveness analysis for incineration or recycling of Dutch household plastic waste, Ecol. Econ. 135 (2017) 22–28.
[46] A. Milbrandt, K. Coney, A. Badgett, G.T. Beckham, Quantification and evaluation of plastic waste in the United States, Resour. Conserv. Recycl. 183 (2022) 106363.
[47] J. Chattopadhyay, T.S. Pathak, R. Srivastava, A.C. Singh, Catalytic co-pyrolysis of paper biomass and plastic mixtures (HDPE (high density polyethylene), PP (polypropylene) and PET (polyethylene terephthalate)) and product analysis, Energy 103 (2016) 513–521.
[48] M. Čolnik, D. Pečar, Ž. Knez, A. Goršek, M. Škerget, Kinetics study of hydrothermal degradation of PET waste into useful products, Processes 10 (1) (2022) 24.
[49] J.M. Encinar, J.F. González, Pyrolysis of synthetic polymers and plastic wastes. Kinetic study, Fuel Process. Technol. 89 (7) (2008) 678–686.
[50] I. Martin-Gullon, M. Esperanza, R. Font, Kinetic model for the pyrolysis and combustion of poly-(ethylene terephthalate) (PET), J. Anal. Appl. Pyrolysis 58–59 (2001) 635–650.
[51] R.K. Singh, B. Ruj, A.K. Sadhukhan, P. Gupta, A TG-FTIR investigation on the co-pyrolysis of the waste HDPE, PP, PS and PET under high heating conditions, J. Energy Inst. 93 (3) (2020) 1020–1035.
[52] M. Dzięcioł, J. Trzeszczyński, Volatile products of poly(ethylene terephthalate) thermal degradation in nitrogen atmosphere, J. Appl. Polym. Sci. 77 (9) (2000) 1894–1901.
[53] S.A. Jabarin, E.A. Lofgren, Thermal stability of polyethylene terephthalate, Polym. Eng. Sci. 24 (13) (1984) 1056–1063.
[54] F. Samperi, C. Puglisi, R. Alicata, G. Montaudo, Thermal degradation of poly (butylene terephthalate) at the processing temperature, Polym. Degrad. Stabil. 83 (1) (2004) 11–17.
[55] M. Dzięcioł, J. Trzeszczyński, Studies of temperature influence on volatile thermal degradation products of poly(ethylene terephthalate), J. Appl. Polym. Sci. 69 (12) (1998) 2377–2381.
[56] V.K. Soni, G. Singh, B.K. Vijayan, A. Chopra, G.S. Kapur, S.S.V. Ramakumar, Thermochemical recycling of waste plastics by pyrolysis: a review, Energy Fuels 35 (16) (2021) 12763–12808.
[57] D. Kwon, S. Jung, D.H. Moon, Y.F. Tsang, W.-H. Chen, E.E. Kwon, Strategic management of harmful chemicals produced from pyrolysis of plastic cup waste using CO_2 as a reaction medium, Chem. Eng. J. 437 (2022) 135524.
[58] S.V. Levchik, E.D. Weil, A review on thermal decomposition and combustion of thermoplastic polyesters, Polym. Adv. Technol. 15 (12) (2004) 691–700.
[59] H. Kurokawa, M.-a. Ohshima, K. Sugiyama, H. Miura, Methanolysis of polyethylene terephthalate (PET) in the presence of aluminium tiisopropoxide catalyst to form dimethyl terephthalate and ethylene glycol, Polym. Degrad. Stabil. 79 (3) (2003) 529–533.
[60] S. Tang, F. Li, J. Liu, B. Guo, Z. Tian, J. Lv, Calcined sodium silicate as solid base catalyst for alcoholysis of poly(ethylene terephthalate), J. Chem. Technol. Biotechnol. 97 (5) (2022) 1305–1314.
[61] R. López-Fonseca, I. Duque-Ingunza, B. de Rivas, S. Arnaiz, J.I. Gutiérrez-Ortiz, Chemical recycling of post-consumer PET wastes by glycolysis in the presence of metal salts, Polym. Degrad. Stabil. 95 (6) (2010) 1022–1028.
[62] S. Tang, F. Li, J. Liu, B. Guo, Z. Tian, J. Lv, MgO/NaY as modified mesoporous catalyst for methanolysis of polyethylene terephthalate wastes, J. Environ. Chem. Eng. 10 (4) (2022) 107927.
[63] Y. Zhang, J. Gao, C. Jiang, G. Luo, J. Fan, J.H. Clark, S. Zhang, Copper-supported catalysts for sustainable PET depolymerization: a cost-effective approach towards dimethyl terephthalate (DMT) production, Green Chem. 26 (11) (2024) 6748–6759.
[64] C.-E. Komly, C. Azzaro-Pantel, A. Hubert, L. Pibouleau, V. Archambault, Multiobjective waste management optimization strategy coupling life cycle assessment and genetic algorithms: application to PET bottles, Resour. Conserv. Recycl. 69 (2012) 66–81.
[65] E. Commission, Packaging and Packaging Waste Regulation, 2023.
[66] Silica Gel Pricing Report, Price Trend, Chart, Market Analysis, News, Demand, Historical and Forecast Data, 2024.